From Words to Meaning: Search, music 🚀 and AI

April 22, 2026

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elastic | The Search AI Company

Elasticsearch

You know, for Search

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APACHE

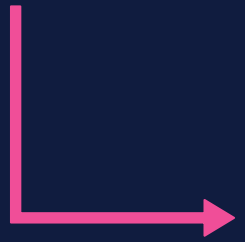
LUCENE™

“These are not the droids you are looking for.”

```
GET /_analyze
{
  "char_filter": [ "html_strip" ],
  "tokenizer": "standard",
  "filter": [ "lowercase", "stop", "snowball" ],
  "text": "These are <em>not</em> the droids
           you are looking for."
}
```

```
"char_filter": "html_strip"
```

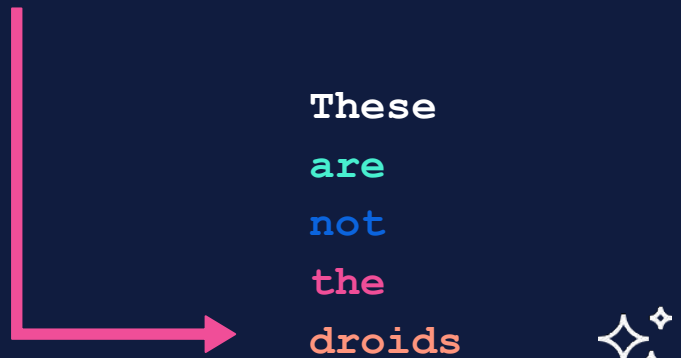
These are not the droids you are looking for.



These are not the droids you are looking for.

"tokenizer": "standard"

These are not the droids you are looking for.



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```
"filter": "lowercase"
```

These	these
are	are
not	not
the	the
droids →	droids
you	you
are	are
looking	looking
for	for

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```
"filter": "stop"
```

These	→	these	→	
are		are		
not		not		
the		the		
droids		droids		droids
you		you		you
are		are		
looking		looking		looking
for		for		

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```
"filter": "snowball"
```

These	these			
are	are			
not	not			
the	the			droid
droids →	droids →	droids →	you	
you	you	you		
are	are			look
looking	looking	looking		
for	for			

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These are not the droids you are looking for.

```
{ "tokens": [{
      "token": "droid",
      "start_offset": 27, "end_offset": 33,
      "type": "<ALPHANUM>", "position": 4
    },{
      "token": "you",
      "start_offset": 34, "end_offset": 37,
      "type": "<ALPHANUM>", "position": 5
    }, {
      "token": "look",
      "start_offset": 42, "end_offset": 49,
      "type": "<ALPHANUM>", "position": 7
    }]}
```

Semantic
search

≠

Literal
matches

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You know, for **Vector** Search

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What is a **Vector** ?

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Embeddings represent your data

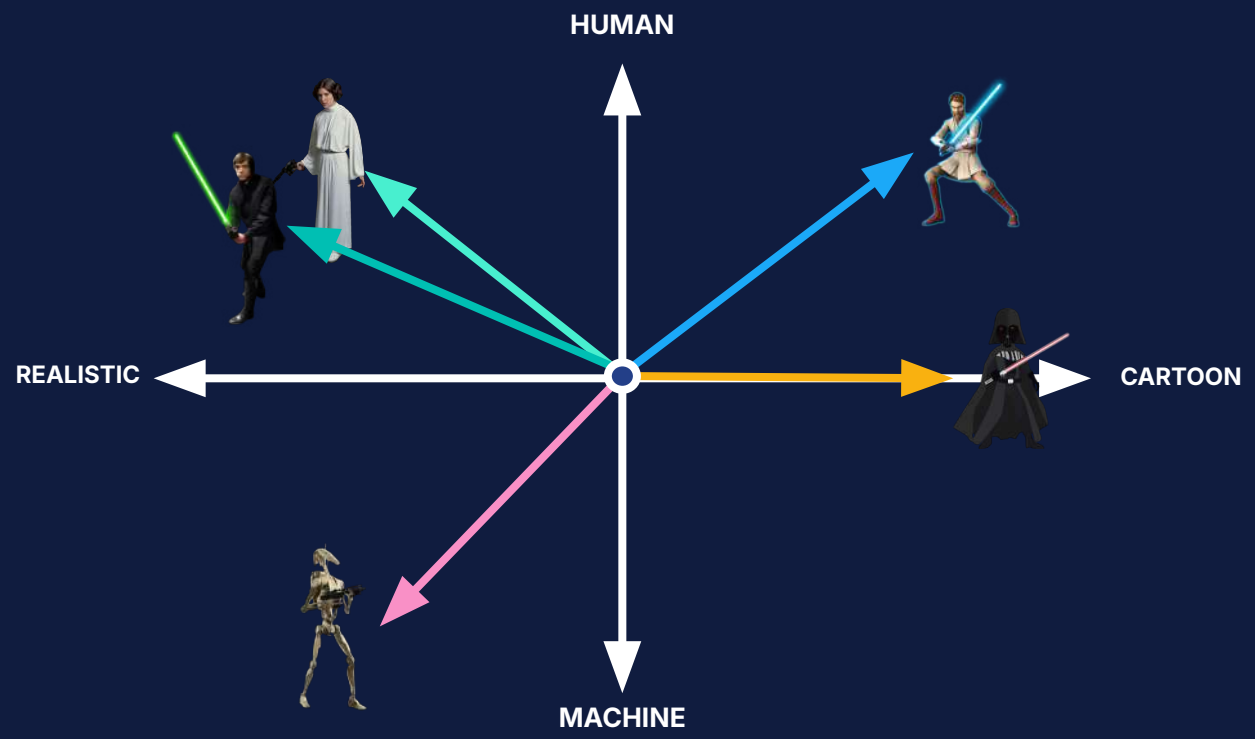
Example: 1-dimensional vector

REALISTIC ← → CARTOON

Character	Vector
	[-1]
	[1]

Multiple dimensions

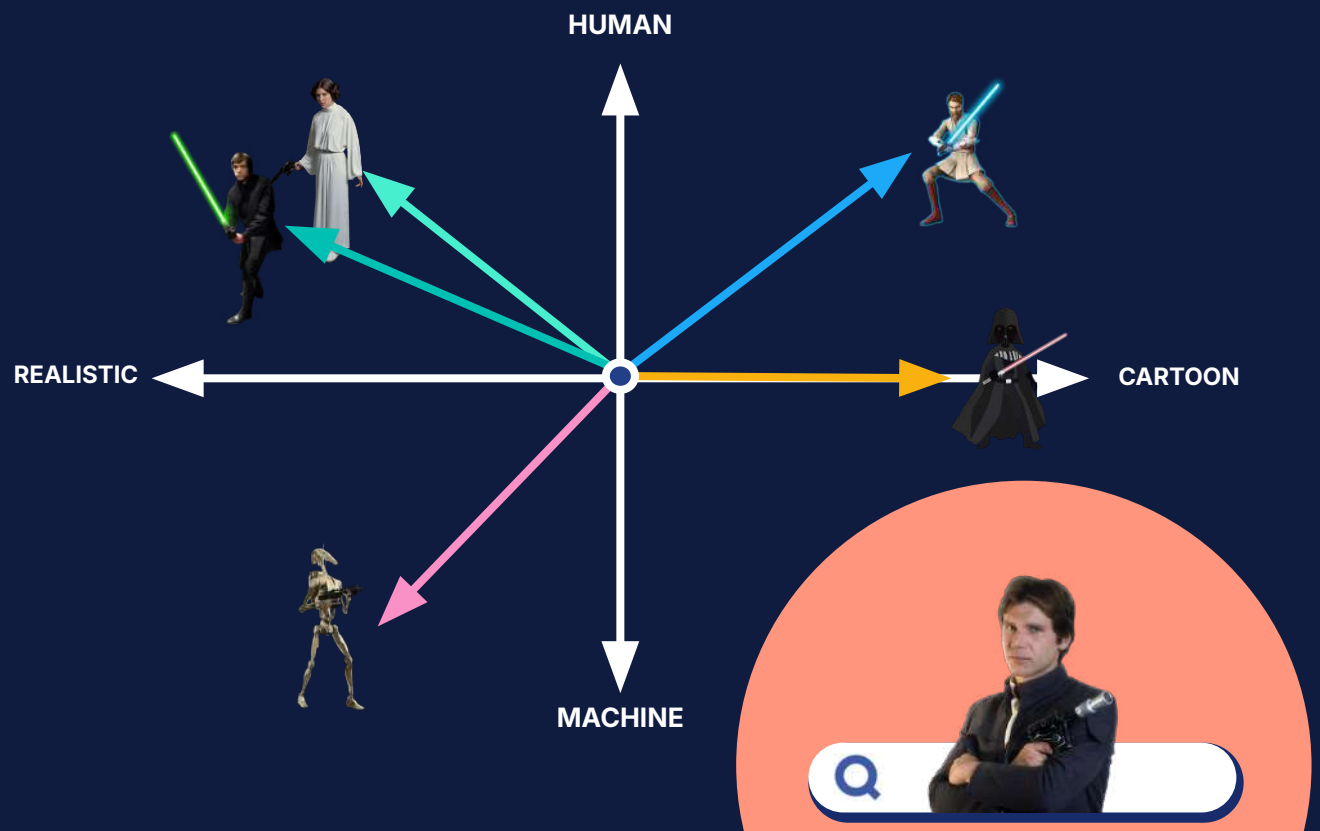
Represent different data aspects



Character	Vector
	[-1.0, 1.0]
	[1.0, 0.0]
	[-1.0, 0.8]
	[1.0, 1.0]
	[-1.0, -1.0]

Vector search ranks objects

By similarity (~relevance) to the query



Relevance	Result
Query	
1	
2	
3	
4	
5	

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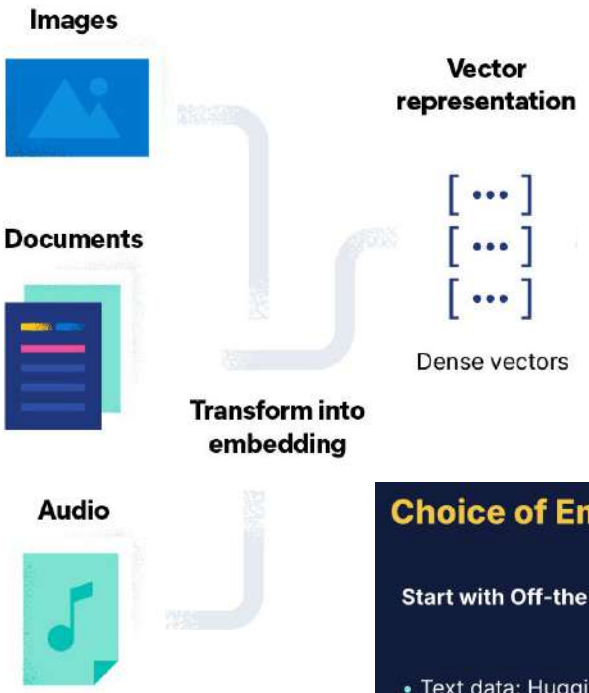
How do you index **Vectors** ?

Vector Search: Architecture



Choice of Embedding Model

Start with Off-the Shelf Models

- Text data: Hugging Face (like Microsoft's E5)
- Images: OpenAI's CLIP

Extend to Higher Relevance

- Apply hybrid scoring
- Bring Your Own Model: requires expertise + labeled data

`dense_vector` field type

```
PUT ecommerce
{
  "mappings": {
    "properties": {
      "description": {
        "type": "text"
      },
      "desc_embedding": {
        "type": "dense_vector"
      }
    }
  }
}
```

Data Ingestion and Embedding Generation

Source data

POST ecommerce/_doc

```
{
  "_id": "product-1234",
  "product_name": "Summer Dress",
  "description": "Our best-selling...",
  "price": 119,
  "color": "blue",
  "fabric": "cotton",
  "desc_embedding": [0.452, 0.3242, …],

}
```

POST ecommerce/_doc

How do you search **Vectors** ?

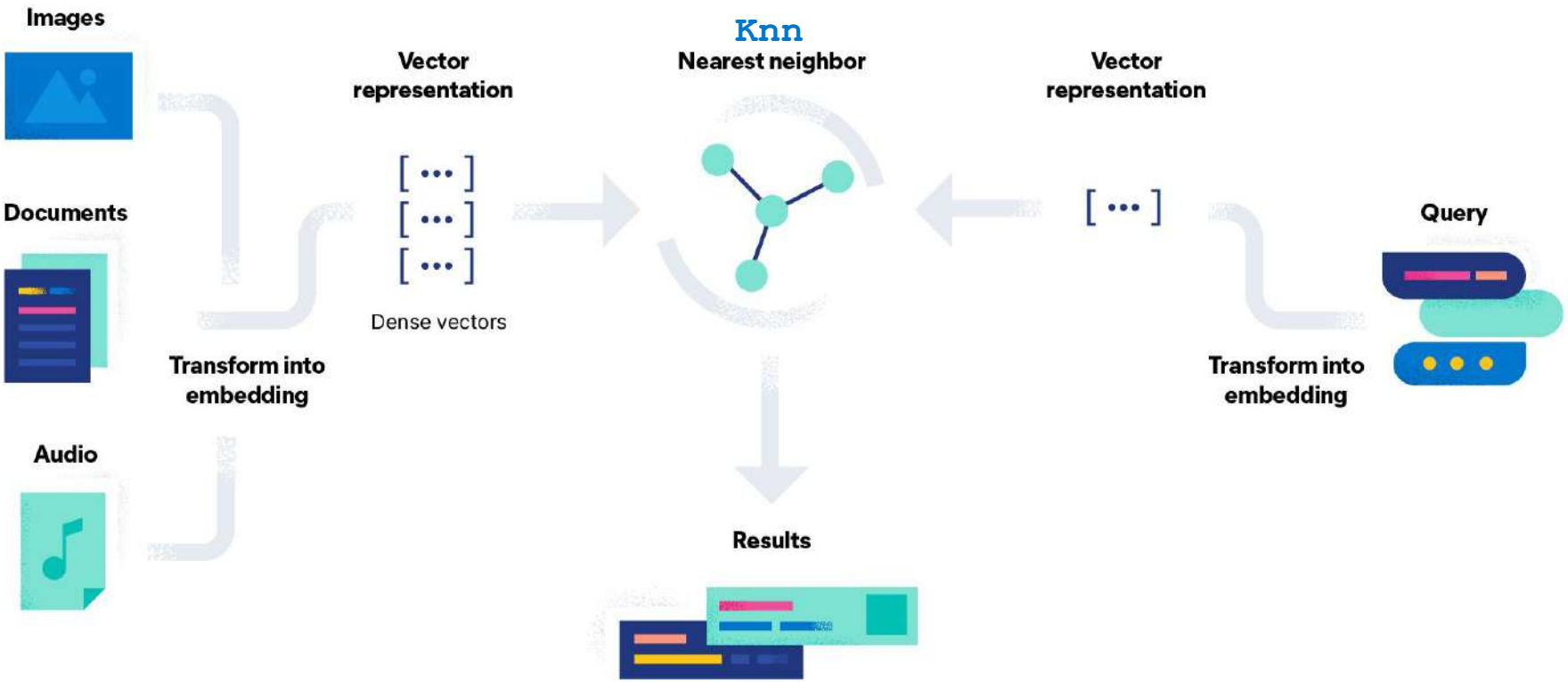
Vector Search: Architecture

Knn query

```
GET ecommerce/_search

{
  "query": {
    "bool": {
      "must": [{
        "knn": {
          "field": "desc_embedding",
          "query_vector": [0.123, 0.244, ...]
          …
        }
      }],
      "filter": {
        "term": {
          "department": "women"
        }
      }
    }
  },
  "size": 10
}
```

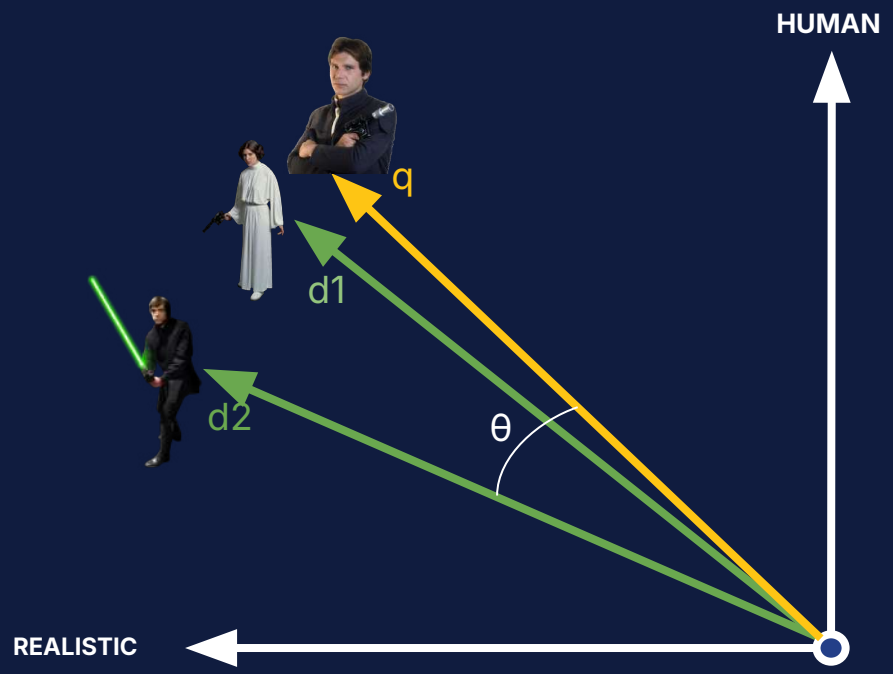
Vector Search: Architecture



But how does it

really work?

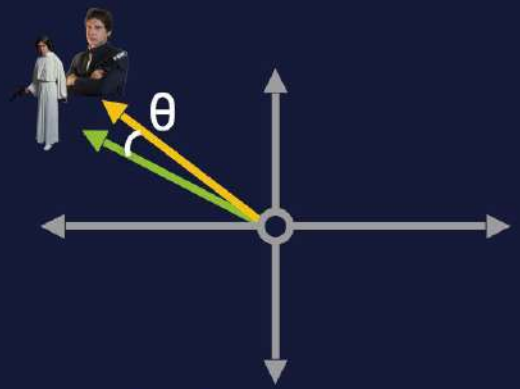
Similarity



$$cos(\theta) = \frac{\vec{q} \times \vec{d}}{|\vec{q}| \times |\vec{d}|}$$

$$_score = \frac{1 + cos(\theta)}{2}$$

Similarity: cosine

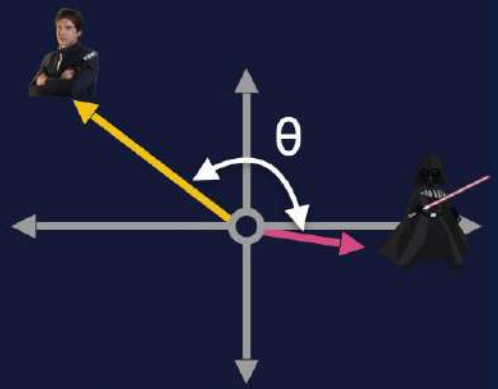


Similar vectors
θ close to 0
cos(θ) close to **1**

$$_score = \frac{1+1}{2} = 1$$

Orthogonal vectors
θ close to 90°
cos(θ) close to **0**

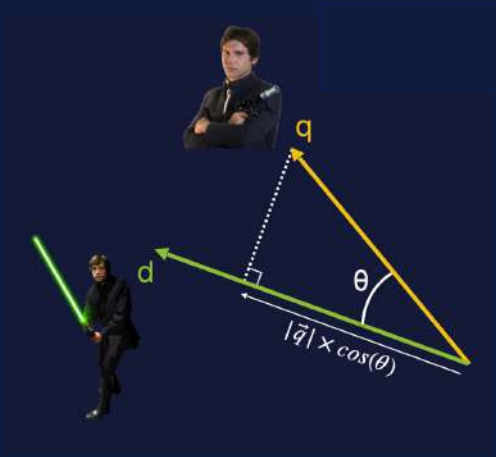
$$_score = \frac{1+0}{2} = 0.5$$



Opposite vectors
θ close to 180°
cos(θ) close to **-1**

$$_score = \frac{1-1}{2} = 0$$

Similarity: Dot product (`dot_product` or `max_inner_product`)

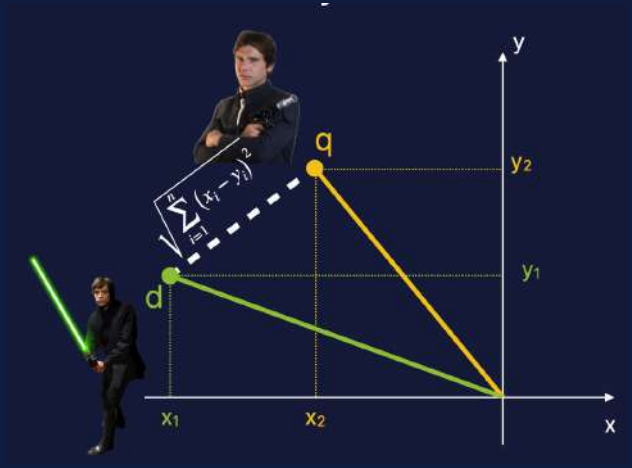


$$\vec{q} \times \vec{d} = |\vec{q}| \times cos(\theta) \times |\vec{d}|$$

$$_score_{float} = \frac{1 + dot_product(q, d)}{2}$$

$$_score_{byte} = \frac{0.5 + dot_product(q, d)}{32768 \times dims}$$

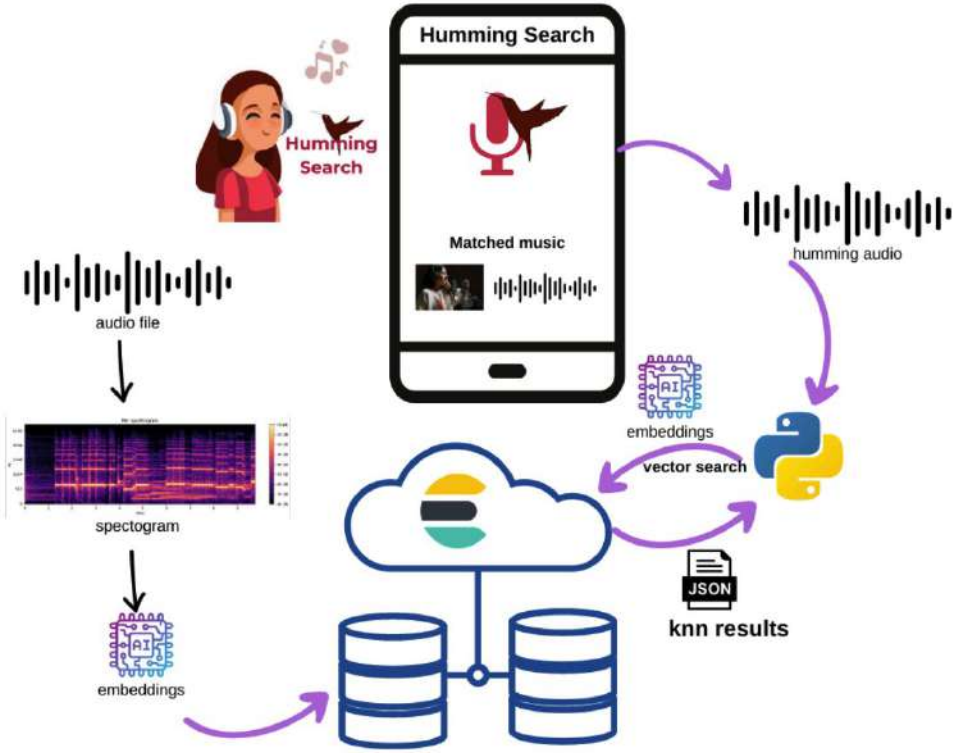
Similarity: Euclidean distance (`l2_norm`)



$$l2_norm_{q,d} = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

$$_score = \frac{1}{1 + (l2_norm_{q,d})^2}$$

Quick Demo



https://github.com/dadoonet/music-search/

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You know, for **Hybrid** Search

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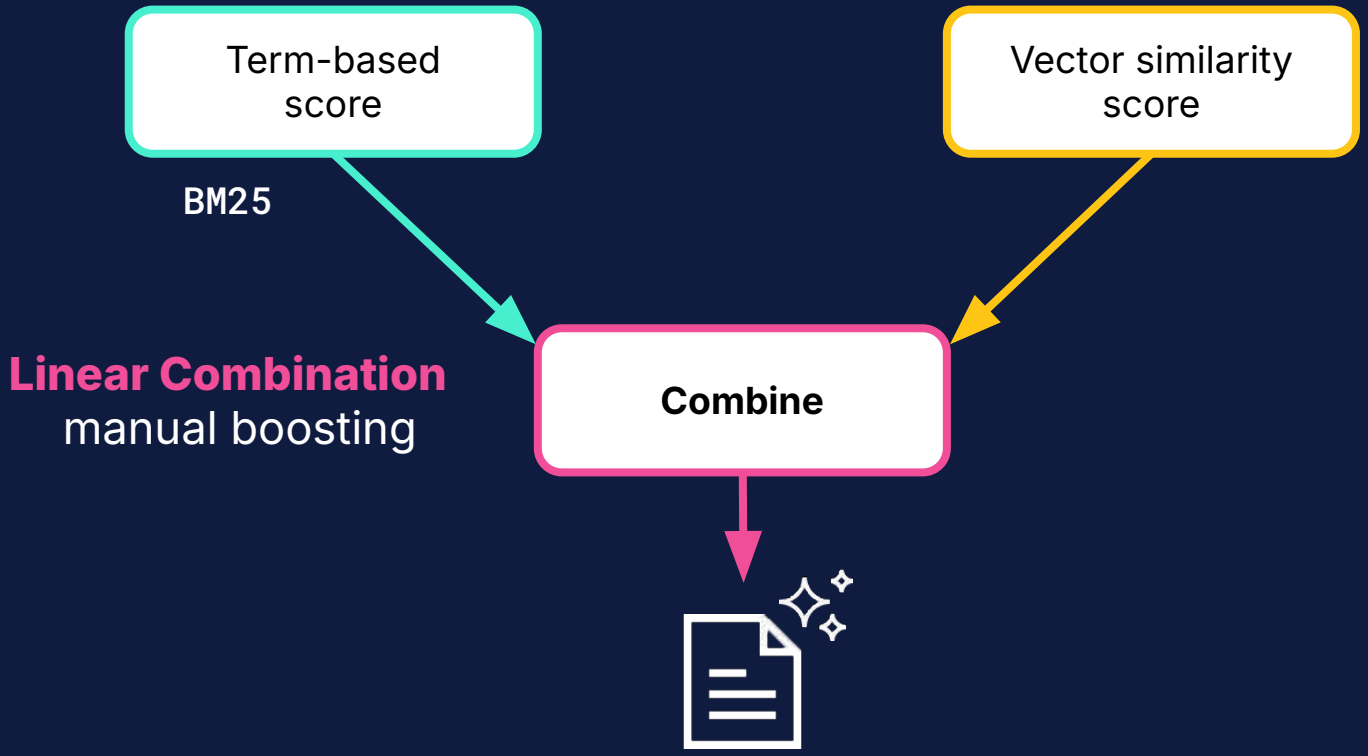
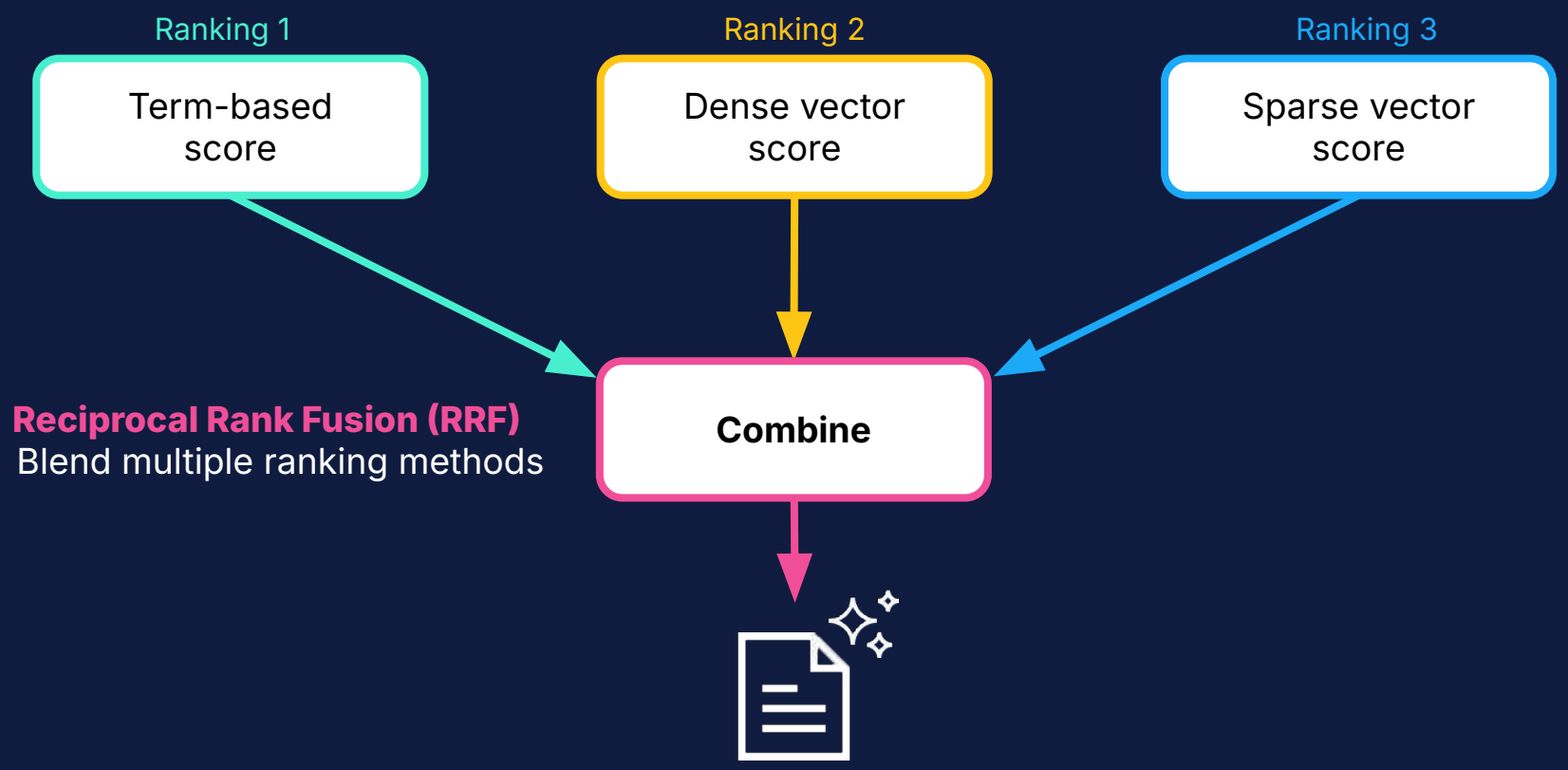
Hybrid scoring



Hybrid ranking



Reciprocal Rank Fusion (RRF)
Blend multiple ranking methods

Reciprocal Rank Fusion (RRF)

$$RRFscore(d \in D) = \sum_{r \in R} \frac{1}{k + r(d)}$$

D - set of docs
R - set of rankings as permutation on 1..|D|
k - typically set to 60 by default

Dense Vector			
Doc	Score	r(d)	k+r(d)
A	1	1	61
B	0.7	2	62
C	0.5	3	63
D	0.2	4	64
E	0.01	5	65

BM25			
Doc	Score	r(d)	k+r(d)
C	1,341	1	61
A	739	2	62
F	732	3	63
G	192	4	64
H	183	5	65

Doc	RRF Score
A	1/61 + 1/62 = 0,0325
C	1/63 + 1/61 = 0,0323
B	1/62 = 0,0161
F	1/63 = 0,0159
D	1/64 = 0,0156

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ChatGPT

Elastic and LLM

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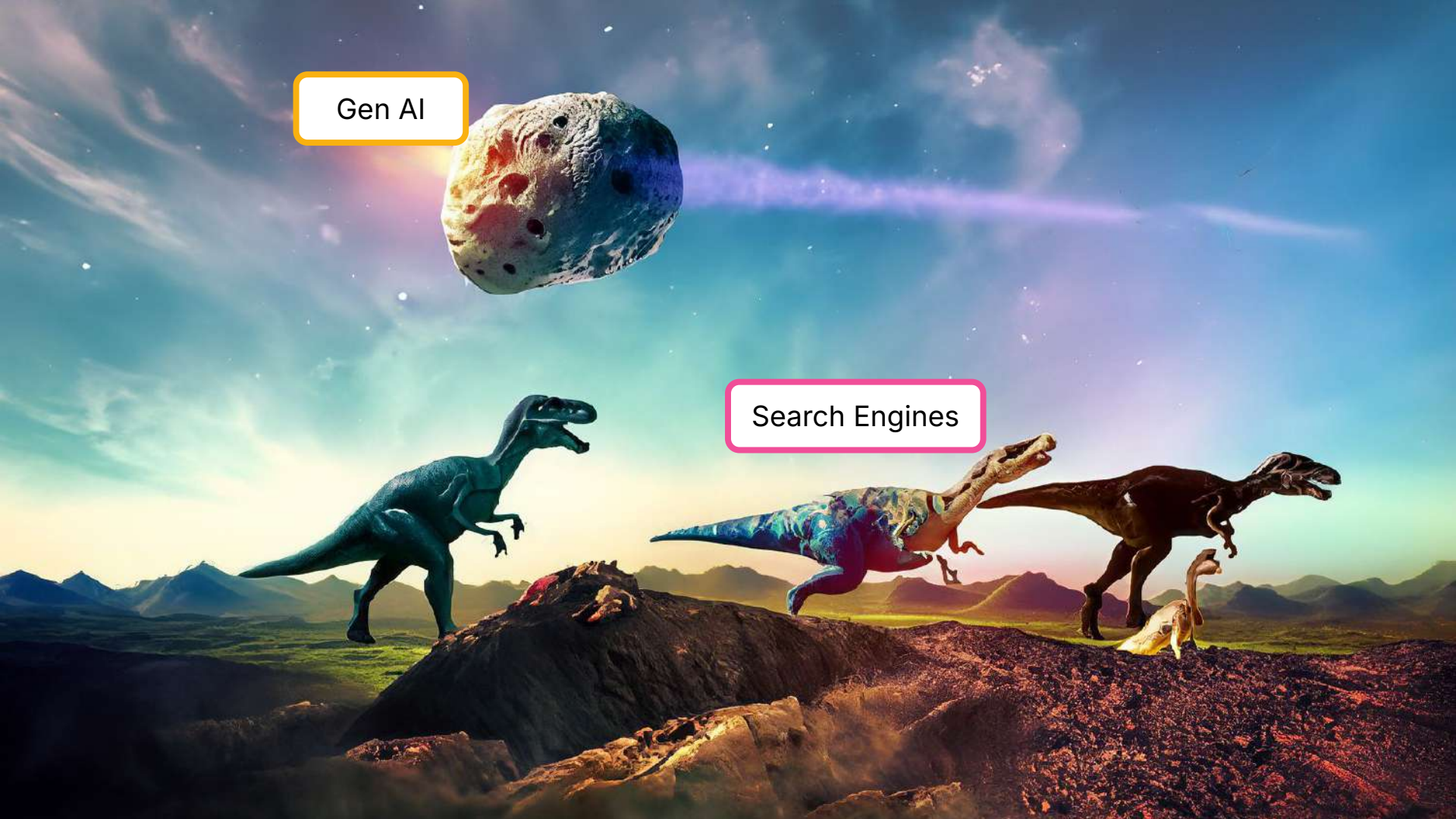
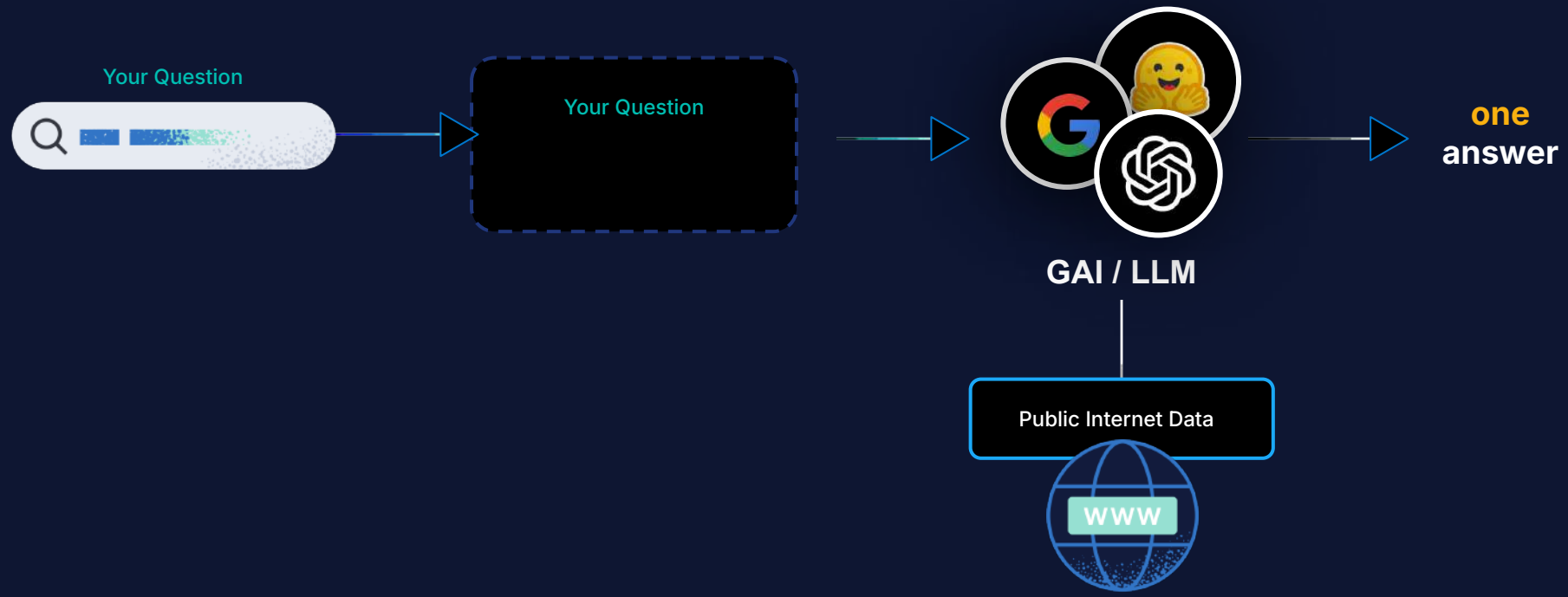

Gen AI
Search Engines

LLM: opportunities and limits



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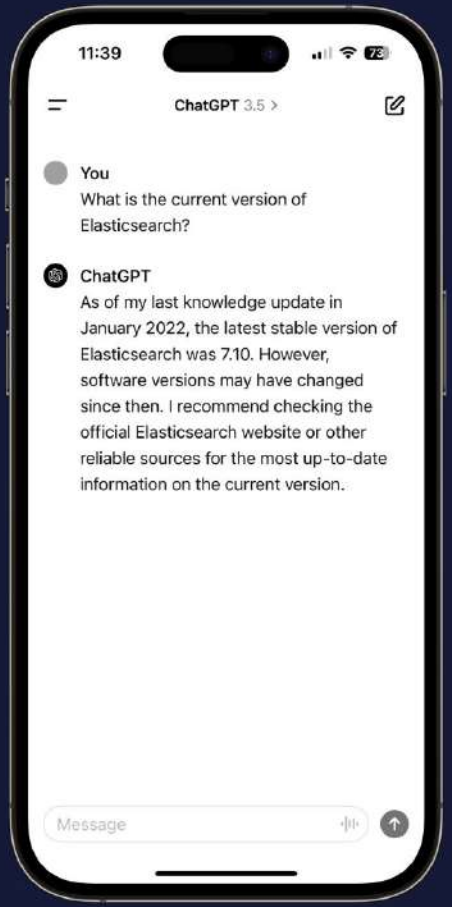
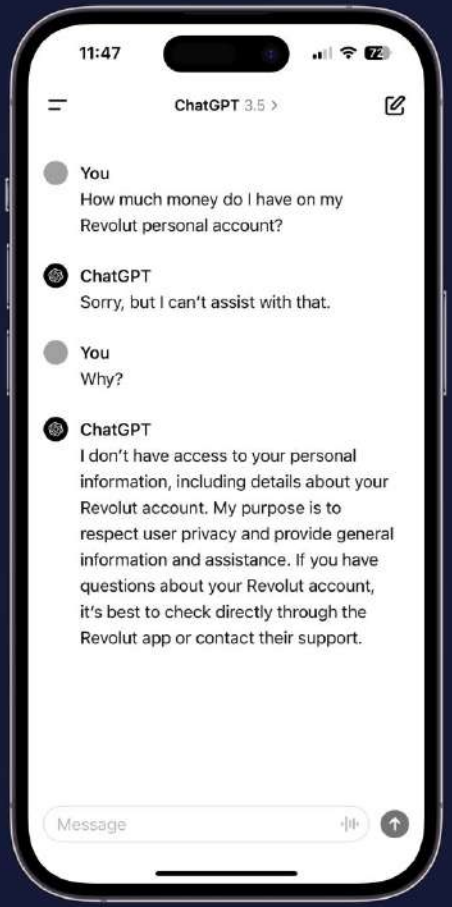
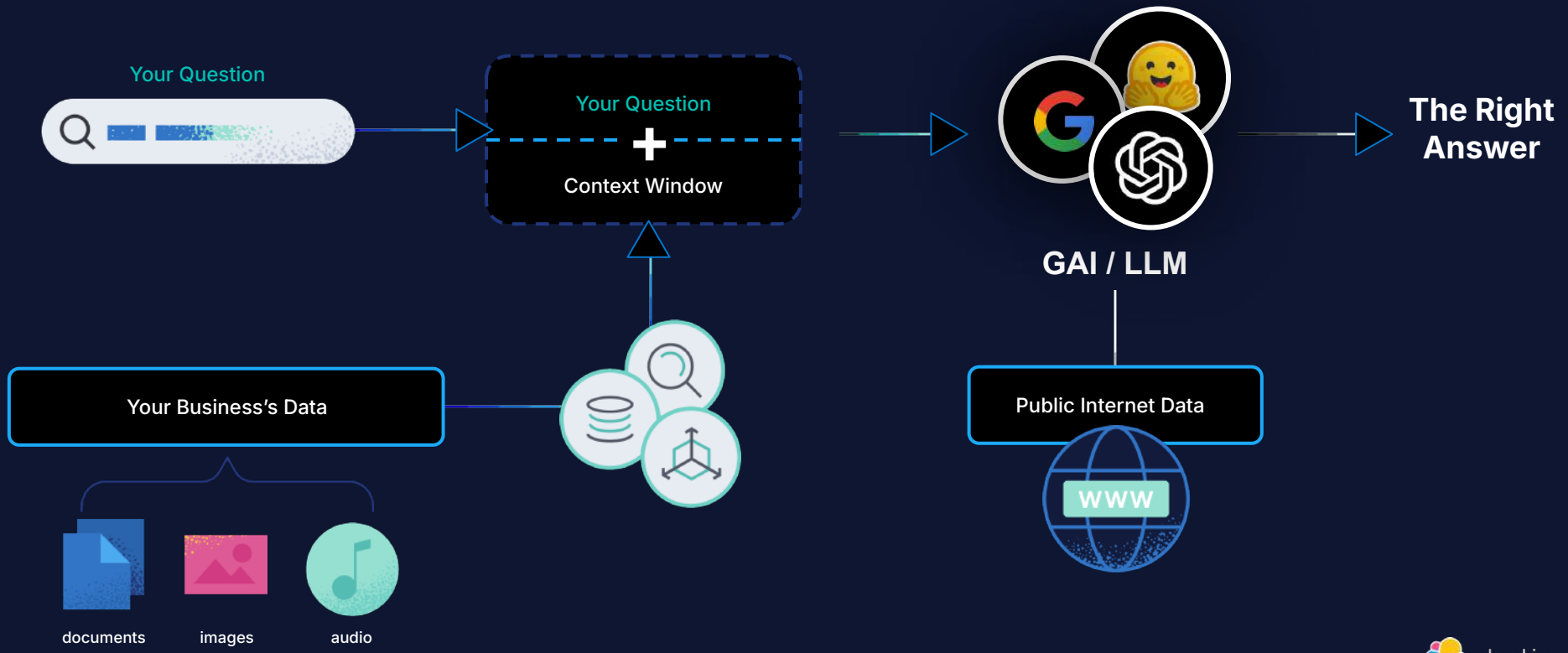

11:39
ChatGPT 3.5 >
You
What is the current version of Elasticsearch?
ChatGPT
As of my last knowledge update in January 2022, the latest stable version of Elasticsearch was 7.10. However, software versions may have changed since then. I recommend checking the official Elasticsearch website or other reliable sources for the most up-to-date information on the current version.
Message


11:47
ChatGPT 3.5 >
You
How much money do I have on my Revolut personal account?
ChatGPT
Sorry, but I can't assist with that.
You
Why?
ChatGPT
I don't have access to your personal information, including details about your Revolut account. My purpose is to respect user privacy and provide general information and assistance. If you have questions about your Revolut account, it's best to check directly through the Revolut app or contact their support.
Message

Retrieval Augmented Generation



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Thank you!

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